Lead Scoring Case Study

Predicting Potential Customers using Logistic Regression

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Course: Advanced Certificate in Data Science

Problem Statement

❑Business Problem

- X Education receives many leads through website visits, forms, referrals, and marketing channels.
- However:

○ Only ~30% leads convert normally

○ Sales team spends time contacting low-quality leads

○ Need to identify **Hot Leads**

❑Objective

- Build a machine learning model to assign lead scores (0–100) and prioritize likely converters.

Dataset Overview

❑Dataset Used

- File: Leads.csv
- Approx. 9000 records
- Target Variable: Converted

❑Key Features

- Lead Source
- Lead Origin
- Total Time Spent on Website
- Total Visits
- Last Activity
- Tags
- Occupation

Data Cleaning & Preparation

❑Steps Performed

- Replaced "Select" values with missing values
- Removed unnecessary columns
- Removed columns with high missing percentage
- Treated missing values
- Created dummy variables for categorical data
- Scaled numeric features
- Split data into train/test sets

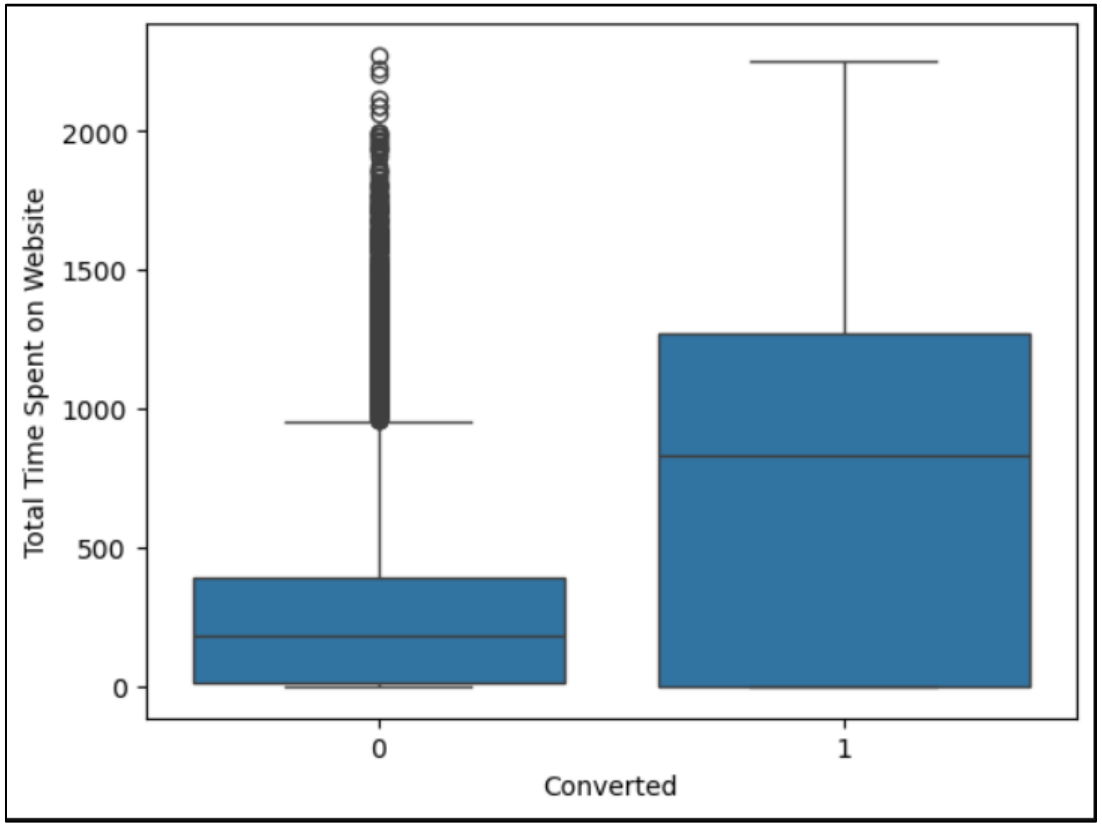
Exploratory Data Analysis

❑Key Insights

- Leads spending more time on website had higher conversion probability
- Leads interacting through forms showed better conversion
- SMS / Phone conversation leads were stronger
- Some lead tags strongly indicated buying intent

❑Examples:

- Countplot of Converted
- Boxplot of Time Spent vs Converted



Model Building

❑Algorithm Used

- Logistic Regression

❑Why Chosen?

- Suitable for binary classification
- Easy to interpret
- Generates probability scores
- Useful for business decision-making

❑Final Feature Selection

- Used statistically significant variables after p-value filtering.

Model Performance

❑Final Results:

- Accuracy = **89.36%**
- ROC AUC Score = **0.94**

❑Confusion Matrix:

[[1521 156]
[139 956]]

Model predicts lead conversion with high reliability.

(Insert ROC curve screenshot)

Top Factors Driving Conversion

❑Most Important Variables

- Tags_Closed by Horizzon
- Tags_Lost to EINS
- Tags_Will revert after reading the email
- Last Notable Activity_SMS Sent
- Total Time Spent on Website

❑Business Meaning

- Leads showing engagement, callbacks, and higher website interest are more likely to convert.

Business Recommendations

❑**Normal Operations:**

- Use threshold = **0.40**
- Prioritize leads with higher lead scores.

❑**Aggressive Conversion Period**

- Use threshold = **0.25 – 0.30**
- Contact more leads to maximize conversions.

❑**Reduce Unnecessary Calls**

- Use threshold = **0.65 – 0.75**
- Contact only highly probable leads.

Conclusion

❑Final Conclusion

- The lead scoring model successfully identifies hot leads and can improve sales efficiency.

❑Benefits to Company

- Better lead prioritization
- Higher conversion rate
- Reduced wasted calls
- Better use of sales team resources

Thank You